A Proof That the Grass Is Greener

Three-shot search on an exponentiated Ornstein–Uhlenbeck landscape

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Abstract

A searcher evaluates a rugged landscape — the exponential of an Ornstein–Uhlenbeck path — at three points chosen sequentially, and is paid the value at the final point. The two-evaluation problem has a three-phase closed form: flee somewhere fresh if the first value is below the median, stay put if it is exceptional, and otherwise step a distance set by the mean-reversion scale. The interesting question is the third evaluation: given two locations a bracket apart, should the searcher ever place the final point between them? A rapidity coordinate $\theta = \operatorname{artanh}(e^{-\kappa\,\Delta t})$, under which bridge means compose additively, reduces the comparison to a hyperbolic identity: an interior bridge point has the same conditional mean as the outside point at the composed rapidity and conditional variance larger by the factor $\cosh(\theta_2+\theta)/\cosh(\theta_2-\theta)$. Since the objective rewards variance, the interior point beats its mean-matched outside rival everywhere except at the bracket's ends. Against the best outside choice the interior wins precisely for anchor values b between a closed-form quadratic root and $e^{2\Theta} = (1+\rho)/(1-\rho)$, with ρ the bracket correlation and Θ its rapidity: stay between the anchors on good news; on bad news step out past the better end, whichever end that is. Within the region the optimal interior point is the bracket middle only up to $b = \sinh 2\tau$, τ the half-bracket rapidity; beyond, a pitchfork sends a mirror pair of optima sliding toward the ends, which they reach exactly at the region's upper edge. Numerically, the optimal three-shot policy widens the bracket relative to the two-shot step in anticipation of the revisit option, which is worth up to about one percent of expected value. In every phase of the policy the winning location is one never yet visited: under a lognormal objective an unvisited point with the same expected height as a visited one is strictly better, by Jensen's inequality, so the grass provably is greener — the searcher's only question is where it is greenest.

1 The problem: three evaluations of an exponentiated OU path

Let X be a stationary Ornstein–Uhlenbeck process on $\mathbb{R}$ with $\mathbb{E}[X_t] = 0$, unit marginal variance, and covariance kernel

$$k(s,t) \;=\; e^{-\kappa|s-t|}. \tag{1}$$

The landscape is $f(t) = \exp(X_t)$: locally continuous, mean-reverting in its logarithm, with occasional high peaks where excursions are magnified by the exponential. A searcher samples f at a first location, which we take to be $t_0 = 0$ by translation invariance, discovering $X_0 = b$. Two further evaluations at locations t_1 and then t_2 follow, each chosen with knowledge of all values so far, and the payoff is the value at the final location,

$$u \;=\; \mathbb{E}[f(t_2)] \;=\; \mathbb{E}\left[e^{X_{t_2}}\right]. \tag{2}$$

Since X_{t_2} is conditionally Gaussian given the observations, with mean μ and variance ν depending on the chosen location,

$$\log \mathbb{E}\left[e^{X_{t_2}}\right] \;=\; \mu + \tfrac{1}{2}\nu, \tag{3}$$

and the searcher is paid for conditional mean and conditional variance in equal measure. This is the engine of everything below: exploration has value not only through what it might reveal along the way but directly, because a final location with residual uncertainty carries a lognormal bonus of half its variance.

Two conventions tidy the boundary cases. A searcher may take a location to infinity, which by mean reversion is equivalent to sampling a fresh $N(0,1)$ value independent of everything seen; and a searcher may revisit an existing location, receiving its known value. Neither changes the substance of any result.

2 Two evaluations: flee, explore, or stay

Given only $X_0 = b$, a candidate second location at distance t has $X_t \sim N\big(b\lambda,\, 1-\lambda^2\big)$ with $\lambda = e^{-\kappa t} \in (0,1]$. From (3),

$$\log \mathbb{E}\left[e^{X_t}\right] \;=\; b\lambda + \tfrac{1}{2}\big(1-\lambda^2\big) \;=\; -\tfrac{1}{2}\left(\lambda - b\right)^2 + \tfrac{1}{2}\left(b^2+1\right), \tag{4}$$

maximized over $\lambda \in [0,1]$ at $\lambda^* = \min\{\max\{b,0\},1\}$. Three phases result.

Proposition 1 (Two-shot policy). The optimal second location and its log-value $\zeta(b)$ are

$$t^* \;=\; \begin{cases} \infty & b<0 \quad \text{(flee)},\\ -\frac{1}{\kappa}\log b & 0 \le b \le 1 \quad \text{(explore)},\\ 0 & b>1 \quad \text{(stay)}, \end{cases} \qquad \zeta(b) \;=\; \begin{cases} \frac{1}{2} & b<0,\\ \frac{1}{2}\left(b^2+1\right) & 0 \le b \le 1,\\ b & b>1. \end{cases} \tag{5}$$

A below-median first value is abandoned entirely; an exceptional one is kept; in between, the searcher steps to the distance at which the prior pull toward the anchor exactly matches the anchor's height. The value ζ will serve as the outside benchmark throughout. Outside the bracket means either ray, and by the Markov property an outside point sees only its nearest anchor, so the best outside move steps past whichever end is better — reversing direction, when the second value disappoints, to leave by the first anchor rather than the second. There is no momentum in the policy; only the heights of the ends matter.

3 Bridge moments and the rapidity coordinate

Now suppose values $X_0 = b$ and $X_{t_1} = d$ are known, and write $\rho = e^{-\kappa t_1}$ for the bracket correlation. For an interior location $t_2 \in (0, t_1)$ let $\lambda_2 = e^{-\kappa t_2}$ and $\lambda = e^{-\kappa (t_1 - t_2)}$ be the correlations to the two ends, so $\lambda_2 \lambda = \rho$. Gaussian conditioning gives the bridge moments

$$\mu \;=\; \frac{\lambda_2\big(1-\lambda^2\big)\, b + \lambda\big(1-\lambda_2^2\big)\, d}{1-\rho^2}, \qquad \nu \;=\; \frac{\big(1-\lambda_2^2\big)\,\big(1-\lambda^2\big)}{1-\rho^2}. \tag{6}$$

The bridge sags: for $b = d > 0$ the mean is below b everywhere strictly inside, pulled toward the process mean, while the variance rises from zero at the ends to a maximum at the middle.

Define rapidities

$$\theta_2 = \operatorname{artanh}(\lambda_2), \qquad \theta = \operatorname{artanh}(\lambda), \tag{7}$$

so named because, in the symmetric case $d = b$, bridge means compose like relativistic velocities:

$$\mu \;=\; b\,\frac{\lambda_2 + \lambda}{1 + \lambda_2 \lambda} \;=\; b \tanh(\theta_2 + \theta)\,. \tag{8}$$

The variance in (6) also takes a hyperbolic product form. Substituting $1 - \tanh^2 = \operatorname{sech}^2$ and using $\cosh(A + B)\cosh(A - B) = \cosh^2\! A \cosh^2\! B - \sinh^2\! A \sinh^2\! B$,

$$\nu \;=\; \frac{\operatorname{sech}^2 \theta_2 \,\operatorname{sech}^2 \theta}{1 - \tanh^2 \theta_2 \tanh^2 \theta} \;=\; \operatorname{sech}(\theta_2 + \theta)\operatorname{sech}(\theta_2 - \theta)\,. \tag{9}$$

An outside location beyond t_1 at correlation $\lambda' = \tanh\theta'$ to its anchor has, by the Markov property, mean $d \tanh\theta'$ and variance

$$1 - \tanh^2 \theta' \;=\; \operatorname{sech}^2 \theta'. \tag{10}$$

4 Interior points beat their mean-matched outside rivals

Equations (8)–(10) settle the comparison between an interior point and the outside point engineered to match its mean.

Theorem 1 (Bridge dominance). Let $d = b$ and let $t_2 \in (0, t_1)$ be an interior location with rapidities (θ_2, θ). The outside location at composed rapidity $\theta' = \theta_2 + \theta$ has the same conditional mean $b \tanh(\theta_2 + \theta)$, and the ratio of conditional variances is

$$\frac{\nu_{\text{inside}}}{\nu_{\text{outside}}} \;=\; \frac{\operatorname{sech}(\theta_2 + \theta)\operatorname{sech}(\theta_2 - \theta)}{\operatorname{sech}^2(\theta_2 + \theta)} \;=\; \frac{\cosh(\theta_2 + \theta)}{\cosh(\theta_2 - \theta)} \;\geq\; 1, \tag{11}$$

with equality only in the degenerate limits $\theta_2 \to \infty$ or $\theta \to \infty$ (the bracket's ends). By (3) the interior point's utility is at least that of its mean-matched outside rival, strictly so in the interior, for every anchor value b.

Proof. The mean statement is (8) against the outside mean $b \tanh\theta'$ at $\theta' = \theta_2 + \theta$; the variance ratio is (9) against (10); and $\cosh(\theta_2 + \theta) \geq \cosh(\theta_2 - \theta)$ with equality iff $\theta_2\theta = 0$. Utility (3) is increasing in ν at fixed μ. $\square$

The interior of a bracket whose two ends are both worth b is, point for point, a better bet than the outside locations with the same expected height: the bridge keeps most of the anchors' mean while retaining, at its middle, variance $\operatorname{sech}(\theta_2 + \theta)$ against the outside's $\operatorname{sech}^2(\theta_2 + \theta)$ — up to three times as much residual uncertainty at the same expectation, and the objective pays for uncertainty.

5 The exact revisit region, and a pitchfork inside it

Dominance over mean-matched rivals does not by itself decide the searcher's move, because the outside player is free to choose any correlation. The relevant comparison is the best interior point against the best outside value ζ of Proposition 1, and it requires knowing where the interior optimum sits. It is not always the middle.

Theorem 2 (Interior critical points, symmetric case). For $d = b > 0$ write the interior log-utility as a function of the correlation $x = \lambda_2 \in (\rho, 1)$ to one end,

$$f(x) \;=\; \frac{b\,(x + \rho/x)}{1+\rho} + \frac{\left(1-x^2\right)\left(1-\rho^2/x^2\right)}{2\,(1-\rho^2)}. \tag{12}$$

Its critical points solve

$$\left(x^2 - \rho\right)\left(x^2 - b\,(1-\rho)\,x + \rho\right) \;=\; 0. \tag{13}$$

The middle $x = \sqrt{\rho}$ is always critical. Writing $\tau = \operatorname{artanh}\sqrt{\rho}$ for the half-bracket rapidity, the middle is the interior optimum if and only if

$$b \;\leq\; \sinh 2\tau \;=\; \frac{2\sqrt{\rho}}{1-\rho}, \tag{14}$$

at which threshold a pitchfork bifurcation occurs: for $b > \sinh 2\tau$ the middle is a local minimum and the optimum is the mirror pair

$$x_{\pm} \;=\; \frac{b\,(1-\rho) \pm \sqrt{b^2\,(1-\rho)^2 - 4\rho}}{2}, \qquad x_+ x_- = \rho, \tag{15}$$

one point seen from each end. The pair is born at the middle and migrates outward, reaching the ends exactly at $b = (1+\rho)\,/\,(1-\rho) = e^{2\Theta}$, where $\Theta = \operatorname{artanh}\rho$ is the full-bracket rapidity.

Proof. The interior problem is secretly one-dimensional in $\sigma = x + \rho/x = \lambda_2 + \lambda$, under which mirror points coincide: using $\left(1-x^2\right)\left(1-\rho^2/x^2\right) = (1+\rho)^2 - \sigma^2$,

$$f \;=\; \frac{b\,\sigma}{1+\rho} + \frac{(1+\rho)^2 - \sigma^2}{2\,(1-\rho^2)}, \tag{16}$$

a downward parabola in σ with peak at $\sigma^* = b\,(1-\rho)$, while σ ranges over $[\,2\sqrt{\rho},\, 1+\rho)$ — its minimum at the middle, its supremum at the ends. Since $d\sigma/dx = 1 - \rho/x^2$ vanishes only at the middle, the critical points of f are the middle together with the preimages of σ^*, which lie in range iff $2\sqrt{\rho} < b\,(1-\rho) < 1+\rho$; solving $x + \rho/x = b\,(1-\rho)$ gives (15) and, with the middle factor, (13). The middle sits at the σ-minimum, so it is the interior optimum iff the parabola is decreasing there, $\sigma^* \leq 2\sqrt{\rho}$, i.e. $b \leq 2\sqrt{\rho}/\,(1-\rho) = 2\tanh\tau\cosh^2\tau = \sinh 2\tau$; beyond, the peak itself is attained by the mirror pair, and $x_+ = 1$ exactly when $\sigma^* = 1+\rho$, i.e. $b = e^{2\Theta}$. □

Corollary 1 (Off-middle value). For $\sinh 2\tau \leq b \leq e^{2\Theta}$ the interior optimum equals the parabola's peak value,

$$U_{\text{off}}(b,\rho) \;=\; \tfrac{1}{2}\left(b^2 e^{-2\Theta} + e^{2\Theta}\right), \tag{17}$$

obtained by substituting $\sigma^* = b\,(1-\rho)$ into (16) and using $e^{2\Theta} = (1+\rho)/(1-\rho)$.

The searcher on very good news does not aim for the center of the bracket: the optimal patch slides from the middle toward the better-known ends as the anchors improve, hugging them ever more closely, and merges with them exactly when b reaches $e^{2\Theta}$.

Theorem 3 (Revisit region, symmetric case). For $d = b \geq 0$ the best interior point strictly beats every outside location precisely for

$$b_-(\rho) \;<\; b \;<\; \frac{1+\rho}{1-\rho} \;=\; e^{2\Theta}, \qquad b_-(\rho) = \frac{\sqrt{\rho}\left(2-\sqrt{2\,(1-\rho)}\right)}{1+\rho}. \tag{18}$$

The lower edge is the smaller root of $b^2\,(1+\rho) - 4b\sqrt{\rho} + 2\rho = 0$, the middle-regime comparison $U_{\mathrm{mid}} = \zeta$ with $U_{\mathrm{mid}}(b,\rho) = 2b\sqrt{\rho}/(1+\rho) + (1-\rho)\,/\,(2\,(1+\rho))$; the upper edge is where the off-middle optima (15) reach the bracket's ends.

Proof. Everything reduces to the parabola (16) and three regimes of its peak. If $b \geq e^{2\Theta}$ then $\sigma^* \geq 1+\rho$: f increases in σ over the whole range, its supremum is the unattained end limit b, and staying wins. If $\sinh 2\tau < b < e^{2\Theta}$ the optimum is U_{off} of Corollary 1, and both comparisons are one line: for $b > 1$,

$$U_{\mathrm{off}} - b \;=\; \tfrac{1}{2}\, e^{-2\Theta}\big(b - e^{2\Theta}\big)^2 \;>\; 0, \tag{19}$$

the arithmetic–geometric mean inequality applied to the two terms of (17), with equality only at the tangency $b = e^{2\Theta}$ that defines the upper edge; and for $b \leq 1$,

$$U_{\mathrm{off}} - \tfrac{1}{2}\big(b^2+1\big) \;=\; \tfrac{1}{2}\big(e^{2\Theta}-1\big)\big(1-b^2e^{-2\Theta}\big) \;>\; 0 \tag{20}$$

since $b \leq 1 < e^{\Theta}$. If $b \leq \sinh 2\tau$ the optimum is the middle. For $b \leq 1$, $U_{\mathrm{mid}} > \zeta$ is the stated quadratic; $b_- \leq 2\sqrt{\rho}/\,(1+\rho) < \sinh 2\tau$ places the lower edge inside this regime, so it is exact. No gap opens above: when $\sinh 2\tau \leq 1$ (i.e. $\rho \leq 3 - 2\sqrt{2}$), $b_+ \geq \sinh 2\tau$ reduces to $(1-\rho)^3 \geq 8\rho^2$, whose left side decreases and right side increases in ρ and which holds at $\rho = 3 - 2\sqrt{2}$; when $\sinh 2\tau > 1$, $b_+ \geq 1$ reduces, with $u = \sqrt{\rho}$, to $2u^2\,(1+u) \geq (1-u)^3$, increasing against decreasing and true at $u = \sqrt{2}-1$ (it reads $20\sqrt{2} \geq 28$); and on $1 < b \leq \sinh 2\tau$, $U_{\mathrm{mid}} \geq b$ reduces to $b \leq (1+u)\,/\,(2\,(1-u))$, which covers the whole stretch because $\sinh 2\tau \leq (1+u)\,/\,(2\,(1-u))$ is $(1-u)^2 \geq 0$. Assembling the regimes, the interior wins exactly on (18). □

Along the two-shot-inherited bracket $\rho = b$ the region opens at $b^* = 0.4233\ldots$, the root of $U_{\mathrm{mid}}(b,b) = \zeta(b)$. Spot values with $\rho = b$: at $b = 0.30$ the middle offers 0.5220 against the outside 0.5450 (go forth); at $b = 0.50$, 0.6381 against 0.6250 (revisit); at $b = 0.90$, 0.9251 against 0.9050 (revisit). By continuity the region extends to $d \neq b$: at $b = 0.5$, $\rho = 0.5$ the interior wins for all $d \geq b$ and loses for $d \leq 0.45$. And notably the region reaches above $b = 1$: even an anchor better than the two-shot stay threshold is worth leaving for a point just inside the bracket, provided $b < e^{2\Theta}$ — wide brackets forgive very good news.

6 The three-shot policy, computed

The first location t_1 should anticipate the endgame. Writing $V(b,d,\rho)$ for the best of the outside values $\zeta(b), \zeta(d)$ and the interior optimum, the three-shot value of a bracket ρ is $\log \mathbb{E}_d\big[e^{V(b,d,\rho)}\big]$ with $d \sim N(b\rho,\,1-\rho^2)$, evaluated by quadrature. Optimizing over ρ yields the policy. Three features, all visible in Table 1 and Figure 1:

First, a below-median start still means flight: for $b < 0$ the optimal bracket correlation tends to zero (the searcher abandons the anchor, collecting fresh draws) and the revisit option

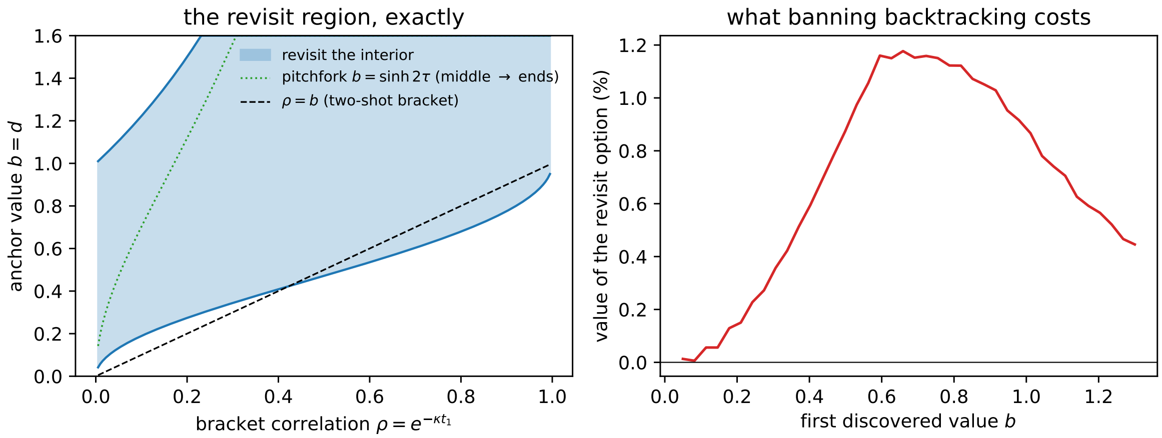

Figure 1: Left: the exact symmetric revisit region (18) in the plane of bracket correlation ρ and anchor value $b = d$, bounded below by the quadratic root b_- and above by $e^{2\Theta} = (1+\rho)\,/\,(1-\rho)$. The dotted curve is the pitchfork $b = \sinh 2\tau$: below it the optimal interior point is the bracket middle, above it the mirror pair (15) hugging the ends. The dashed line is the two-shot bracket $\rho = b$, entering the region at $b^* = 0.4233$. Right: the value of the revisit option — the percentage increase in expected payoff of the full three-shot optimum over the best policy forbidden to backtrack — as a function of the first discovered value b, with the bracket optimized in both cases.

is worthless, since a bridge between an abandoned anchor and whatever follows is a bridge between things not worth bridging.

Second, the optimal bracket is wider than the two-shot step. At $b = 0.5$ the two-shot policy steps to $\rho = 0.5$; the three-shot optimum stretches to $\rho = 0.28$. The wide bracket is bought in anticipation of the endgame: a wider bracket makes the eventual bridge middle more uncertain, and (3) pays half of that uncertainty directly.

Third, the revisit option's worth peaks near one percent of expected value around $b \approx 0.7$ and vanishes at both extremes — worthless when the start is abandoned, nearly worthless when the start is so good the policy barely moves. Forbidding backtracking is thus a cheap approximation by value, but it is not the optimal policy, and its prescription is qualitatively wrong exactly when both evaluations came in high.

7 Discussion

The proverb, read as mathematics, is Jensen's inequality: under a payoff that exponentiates a Gaussian, any unvisited location whose expected height matches a visited one is strictly better, carrying a bonus of half its residual variance. The grass is greener — provably, not wistfully — and the analysis above locates the greenest patch. On bad news, leave: a below-median anchor is abandoned outright, and a disappointing second value sends the final point out past the better end — in whichever direction that end lies. On good news, stay between: when both ends of the bracket came in high, the sagging bridge between them retains most of their promise and, at its middle, several times the residual uncertainty of any outside point with the

b	optimal ρ	value	value, no revisit	revisit worth
-0.5	$\to 0$	0.8306	0.8306	0.00%
0.1	0.06	0.8398	0.8394	0.03%
0.3	0.16	0.8728	0.8695	0.33%
0.5	0.28	0.9392	0.9304	0.88%
0.7	0.40	1.0378	1.0262	1.16%
0.9	0.50	1.1670	1.1567	1.03%
1.2	0.62	1.4060	1.4004	0.56%

Table 1: The three-shot game: optimal bracket correlation, log-value, the log-value of the best policy forbidden to revisit the bracket interior, and the percentage the revisit option adds to expected payoff. Gauss–Hermite quadrature with 61 nodes over the second value d; interior optima on a 400-point grid.

same expectation — and an objective that exponentiates a Gaussian pays for uncertainty at half-weight. The exact frontier between the two regimes is the quadratic (**??**).

A caution about the word exploration. With one evaluation remaining there is nothing to explore: no discovery at the final point can inform a later choice, because there is none. The last move is pure placement — put the point where $\mu + \frac{1}{2}\nu$ is largest — and the pull toward uncertain locations is Jensen's inequality, not curiosity: the objective pays half the variance directly, whether or not anything is ever learned there. Exploration proper enters only with more shots remaining, where early placements buy information. The natural conjecture from the endgame here is a two-phase k-shot policy: step out past the running best while values keep climbing, each new high re-anchoring the next step, then refine between the two highest once a good region is bracketed. Settling it is a dynamic program over the posterior, cheap per step with the bridge formulas above.

The rapidity coordinate is the tool worth keeping: bridge means compose additively in $\theta = \operatorname{artanh}(e^{-\kappa\Delta t})$, bridge variances factor as $\operatorname{sech}(\theta_2 + \theta)\operatorname{sech}(\theta_2 - \theta)$, and the inside–outside comparison collapses to the monotonicity of cosh. The same coordinate is the natural discretization for lattice computations of extremes of OU paths, where uniform-in-θ spacing makes correlations additive across cells.

Beyond three shots the closed forms give out, and the k-shot policy is a dynamic program over the posterior — bridge moments per interval, maximization over intervals — that the formulas here make cheap per step. The landscapes themselves remain a useful synthetic test bed for few-shot exploration algorithms: the choice between striking out and revisiting is a one-bit probe of whether a search strategy has grasped the trade between mean reversion and the variance bonus, and (**??**) is the answer key.

Reproducibility

All numbers and the figure are produced by `numerics.py` alongside this source, with an independent verification of Theorems 1 and 3 by exact conditioning and weighted Monte Carlo in `verify/check_inside.py`, in the browniansearch repository.

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